

ANGERMUENDE, Germany

WMO: 102910

Lat: 53.0317N

Lon: 13.9908E

Elev: 55

StdP: 100.66

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.1	-9.7	-15.7	1.0	-12.1	-12.2	1.3	-9.4	13.4	7.3	12.0	5.3	2.2	70	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	29.1	19.3	27.0	18.4	25.3	17.7	20.4	26.9	19.3	25.0	18.3	23.6	3.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.2	22.1	17.2	12.4	21.4	16.2	11.6	20.8	58.8	27.1	55.2	25.0	52.0	23.5	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.8	8.6	-14.2	32.9	6.3	2.5	-18.7	34.7	-22.3	36.2	-25.8	37.6	-30.4	39.4		
			-14.5	22.1	6.1	1.0	-18.9	22.8	-22.5	23.4	-25.9	24.0	-30.3	24.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	0.6	0.0	4.0	7.7	12.6	15.8	17.9	17.9	14.0	9.3	4.0	1.6
	DBStd	7.48	5.49	4.97	3.99	3.59	3.40	3.17	2.95	3.17	2.77	3.72	3.53	3.94
	HDD10.0	1360	292	279	188	88	13	0	0	0	2	56	181	261
	HDD18.3	3546	550	512	445	319	179	89	44	48	132	279	431	518
	CDD10.0	939	0	1	2	18	94	174	246	244	123	36	1	0
	CDD18.3	83	0	0	0	0	2	13	31	34	3	0	0	0
	CDH23.3	862	0	0	0	2	33	146	328	317	35	1	0	0
	CDH26.7	213	0	0	0	0	1	31	81	98	3	0	0	0
Wind	WSAvg	4.0	5.1	4.2	4.3	3.8	3.4	3.4	3.4	3.3	3.7	3.9	4.4	4.8
Precipitation	PrecAvg	554	42	34	37	29	53	59	68	61	50	38	40	43
	PrecMax	762	89	82	71	67	113	149	218	164	122	89	100	82
	PrecMin	408	0	5	11	2	15	14	14	25	14	12	3	5
	PrecStd	81	22	17	18	15	24	33	42	27	23	21	19	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	13.7	16.8	22.3	26.0	30.1	31.6	32.8	26.6	21.1	13.1	11.7
		MCWB	9.1	9.8	10.3	13.0	16.9	18.6	20.5	21.1	17.9	15.8	10.5	9.2
	2%	DB	9.2	10.6	13.3	18.9	24.2	27.3	28.7	29.4	24.0	18.3	11.5	10.4
		MCWB	7.5	8.0	9.0	12.0	15.6	18.0	19.5	19.5	17.0	14.2	9.8	8.7
	5%	DB	8.1	8.0	11.0	16.6	22.3	25.0	27.0	26.7	21.6	16.3	10.1	9.1
		MCWB	6.5	5.8	7.7	11.0	14.8	16.9	18.6	18.4	15.7	13.0	8.6	7.7
	10%	DB	6.8	6.0	9.4	14.2	20.1	22.8	25.0	24.4	19.3	14.6	8.8	7.6
		MCWB	5.3	4.3	6.8	9.6	13.8	16.0	18.1	17.5	14.6	11.8	7.6	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	10.4	11.2	13.8	18.1	20.9	22.0	21.6	19.2	16.7	11.3	9.7
		MCDB	10.8	12.9	15.7	20.1	23.9	25.3	28.3	30.7	25.1	20.1	12.2	10.9
	2%	WB	7.8	7.9	9.5	12.5	16.6	19.0	20.5	20.2	17.7	14.9	10.0	8.9
		MCDB	9.0	10.0	12.5	18.1	22.4	25.0	27.0	26.9	22.1	17.7	11.2	10.2
	5%	WB	6.6	6.1	8.1	11.2	15.3	17.7	19.4	19.2	16.5	13.5	8.7	7.7
		MCDB	8.0	7.9	10.4	15.7	21.2	23.3	25.2	24.7	20.5	15.7	9.8	8.9
	10%	WB	5.4	4.4	6.8	10.0	14.2	16.6	18.3	18.2	15.4	12.2	7.7	6.3
		MCDB	6.8	5.8	9.2	13.9	19.4	21.8	23.6	23.3	18.6	14.1	8.7	7.6
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.7	6.9	9.0	10.2	9.6	10.0	9.9	8.9	7.4	4.9	4.1
		MCDBR	4.7	7.1	9.9	12.5	14.4	14.1	14.5	14.3	12.7	9.6	5.8	5.4
		MCWBR	4.2	5.6	6.6	7.2	7.8	6.8	6.8	6.3	6.9	6.1	4.8	4.9
	5% WB	MCDWR	4.8	7.3	8.9	11.7	12.8	11.6	12.8	11.9	10.9	8.7	5.7	5.3
		MCWBR	4.5	5.9	6.4	7.0	7.5	6.3	6.6	6.0	6.6	6.0	4.9	5.0
Clear-Sky Solar Irradiance	taub	0.317	0.343	0.378	0.412	0.409	0.407	0.427	0.421	0.384	0.366	0.342	0.312	
	taud	2.319	2.307	2.238	2.210	2.260	2.293	2.250	2.279	2.359	2.375	2.366	2.327	
	Ebn at Noon	654	752	796	817	841	846	818	800	786	716	619	588	
	Edh at Noon	61	83	109	127	127	124	127	117	96	78	59	51	
All-Sky Solar Radiation	RadAvg	0.61	1.29	2.42	4.05	5.02	5.30	4.95	4.24	3.09	1.67	0.73	0.43	
	RadStd	0.07	0.20	0.29	0.56	0.60	0.42	0.66	0.36	0.35	0.22	0.10	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page